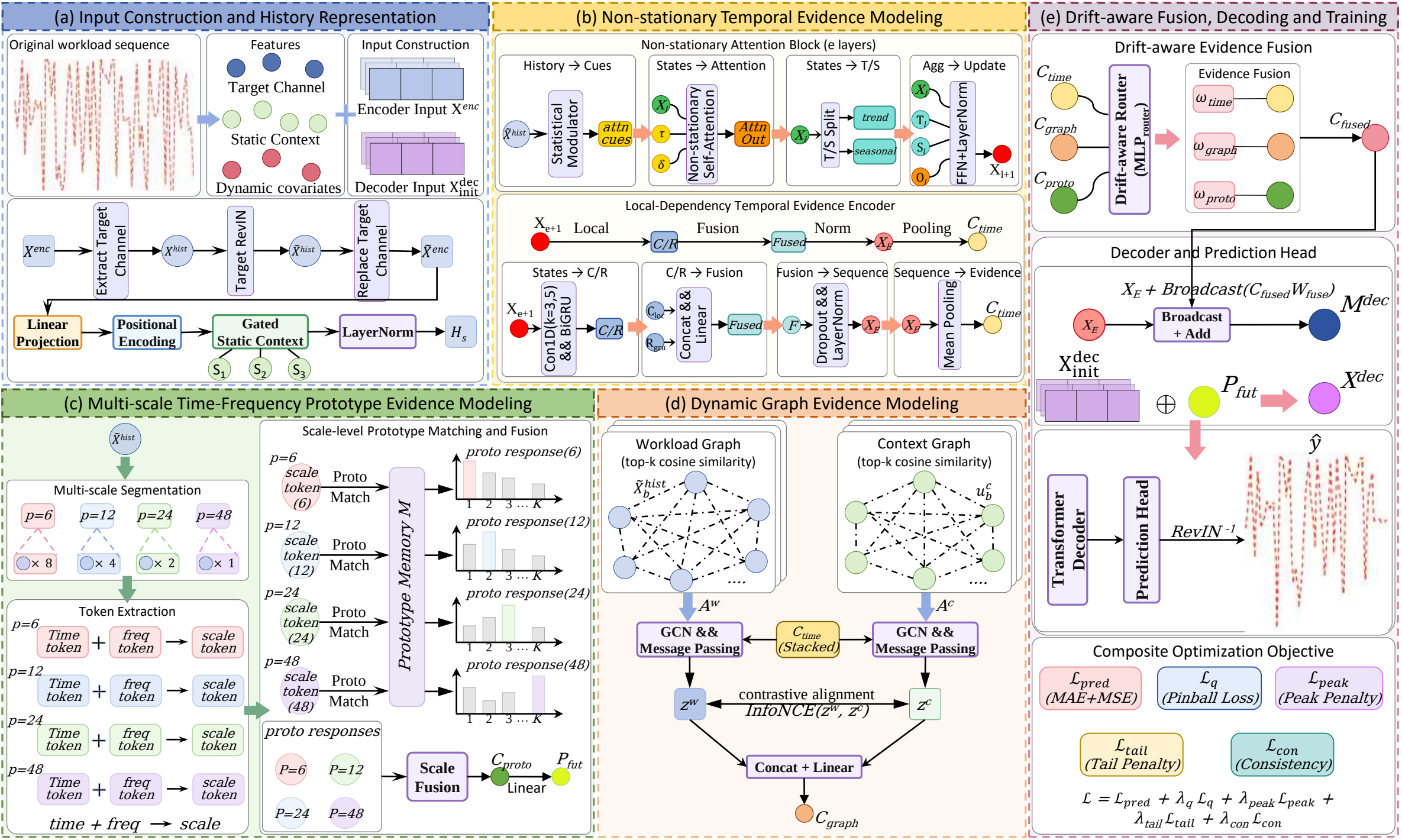

(a) Input Construction and History Representation
Original workload sequence
Features
Target Channel
Static Context
Dynamic covariates
Input Construction
Encoder Input X^enc
Decoder Input X^dec_init
Extract Target Channel
Target RevIN
Replace Target Channel
Linear Projection
Positional Encoding
Gated Static Context
LayerNorm
(b) Non-stationary Temporal Evidence Modeling
Non-stationary Attention Block (e layers)
History → Cues
Statistical Modulator
attn cues
States → Attention
Non-stationary Self-Attention
Attn Out
States → T/S
T/S Split
trend
seasonal
Agg → Update
FFN+LayerNorm
Local-Dependency Temporal Evidence Encoder
Local
Fusion
Norm
Pooling
States → C/R
Con1D(k=3,5) && BiGRU
C/R → Fusion
Concat && Linear
Fusion → Sequence
Dropout && LayerNorm
Sequence → Evidence
Mean Pooling
(c) Multi-scale Time-Frequency Prototype Evidence Modeling
Multi-scale Segmentation
Token Extraction
time + freq → scale
Scale-level Prototype Matching and Fusion
Proto Match
Prototype Memory M
proto responses
Scale Fusion
Linear
(d) Dynamic Graph Evidence Modeling
Workload Graph
(top-k cosine similarity)
Context Graph
(top-k cosine similarity)
GCN && Message Passing
C_time (Stacked)
contrastive alignment
InfoNCE(z^w, z^c)
Concat + Linear
(e) Drift-aware Fusion, Decoding and Training
Drift-aware Evidence Fusion
Drift-aware Router (MLP_router)
Evidence Fusion
Decoder and Prediction Head
Broadcast + Add
Transformer Decoder
Prediction Head
RevIN^-1
Composite Optimization Objective
L_pred (MAE+MSE)
L_q (Pinball Loss)
L_peak (Peak Penalty)
L_tail (Tail Penalty)
L_con (Consistency)